

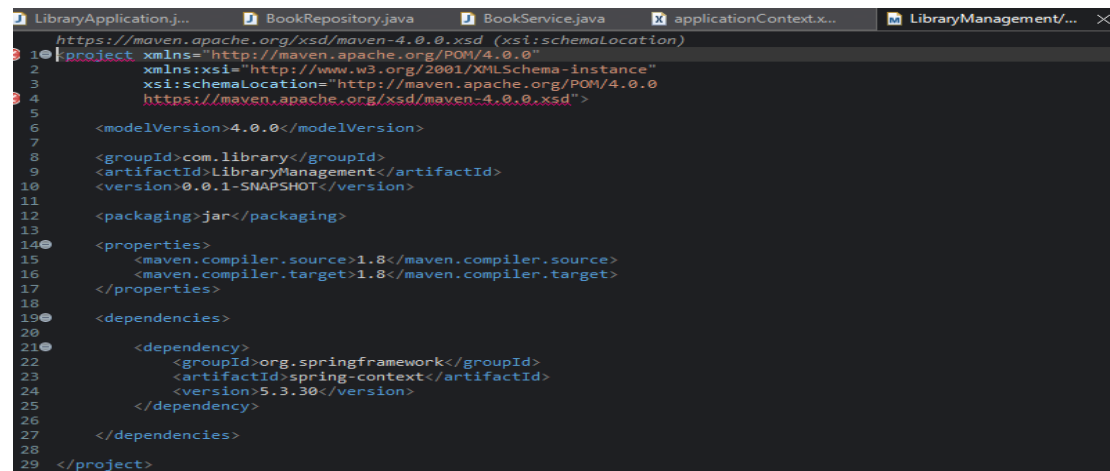
Spring Core Maven

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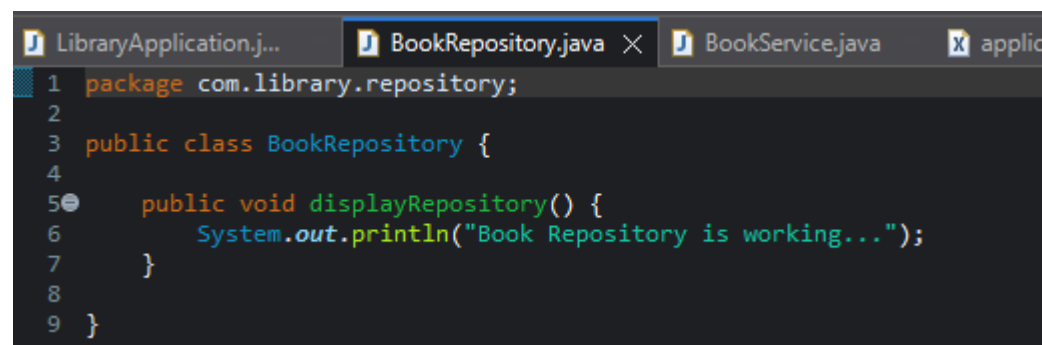
Exercise 1: Configuring a Basic Spring Application

LibraryManagementPom.xml:



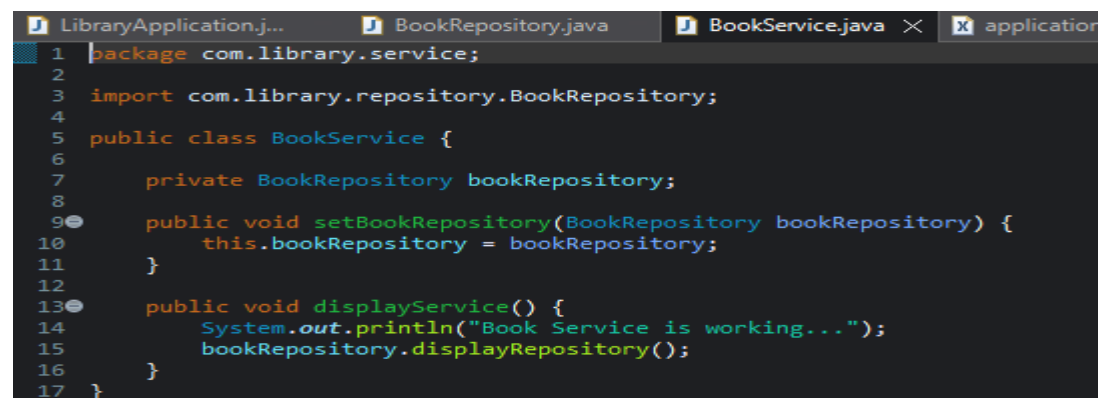
```
1 <?xml version="1.0" encoding="UTF-8" ?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0"
3         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
5                             https://maven.apache.org/xsd/maven-4.0.0.xsd">
6     <modelVersion>4.0.0</modelVersion>
7     <groupId>com.library</groupId>
8     <artifactId>LibraryManagement</artifactId>
9     <version>0.0.1-SNAPSHOT</version>
10    <packaging>jar</packaging>
11
12    <properties>
13        <maven.compiler.source>1.8</maven.compiler.source>
14        <maven.compiler.target>1.8</maven.compiler.target>
15    </properties>
16
17    <dependencies>
18        <dependency>
19            <groupId>org.springframework</groupId>
20            <artifactId>spring-context</artifactId>
21            <version>5.3.30</version>
22        </dependency>
23        <dependency>
24            <groupId>org.springframework</groupId>
25            <artifactId>spring-framework</artifactId>
26            <version>4.0.0</version>
27        </dependency>
28    </dependencies>
29 </project>
```

BookRepository.java:



```
1 package com.library.repository;
2
3 public class BookRepository {
4
5     public void displayRepository() {
6         System.out.println("Book Repository is working...");
7     }
8
9 }
```

BookService.java:



```
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6
7     private BookRepository bookRepository;
8
9     public void setBookRepository(BookRepository bookRepository) {
10         this.bookRepository = bookRepository;
11     }
12
13     public void displayService() {
14         System.out.println("Book Service is working...");
15         bookRepository.displayRepository();
16     }
17 }
```

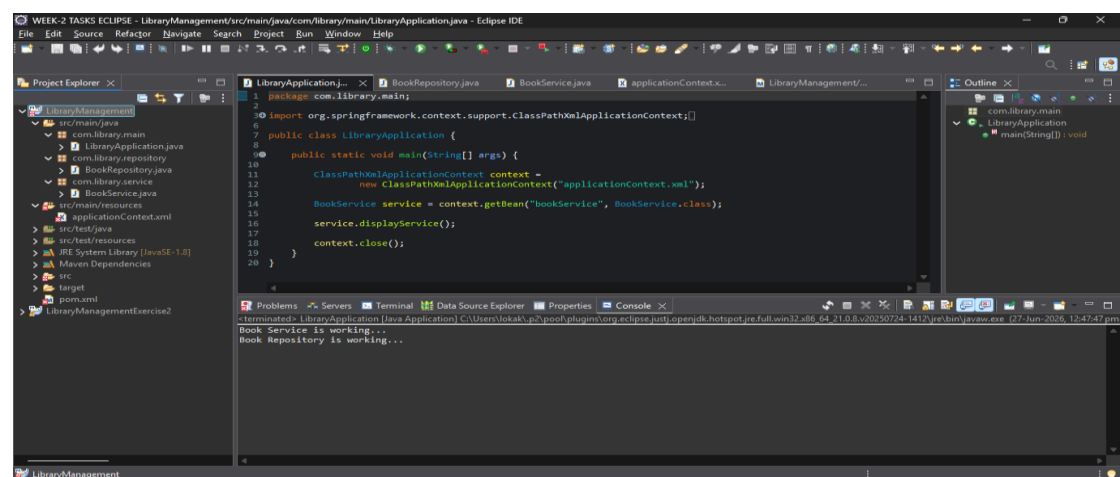
ApplicationContext.xml:

```
LibraryApplication.j... BookRepository.java BookService.java applicationContext.x... X Libra
https://www.springframework.org/schema/beans/spring-beans.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="bookRepository"
10          class="com.library.repository.BookRepository"/>
11
12     <bean id="bookService"
13          class="com.library.service.BookService">
14
15         <property name="bookRepository"
16                   ref="bookRepository"/>
17
18     </bean>
19
20 </beans>
```

LibraryApplication.java:

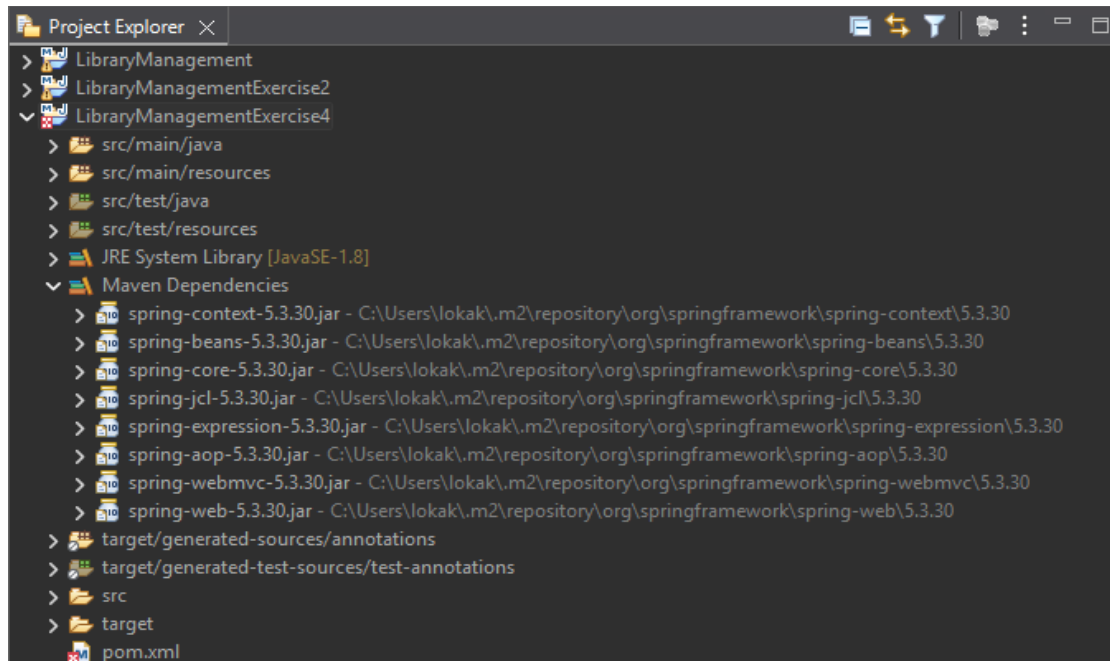
```
LibraryApplication.j... X BookRepository.java BookService.java applicationContext.x... M Libra
1 package com.library.main;
2
3 import org.springframework.context.support.ClassPathXmlApplicationContext;
4
5
6
7 public class LibraryApplication {
8
9     public static void main(String[] args) {
10
11         ClassPathXmlApplicationContext context =
12             new ClassPathXmlApplicationContext("applicationContext.xml");
13
14         BookService service = context.getBean("bookService", BookService.class);
15
16         service.displayService();
17
18         context.close();
19     }
20 }
```

Output:



Exercise 4: Creating and Configuring a Maven Project:

Maven Dependencies:



LibraryManagementExercise4Pom.xml:

LibraryManagementExercise4/pom.xml X

```
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <project xmlns="http://maven.apache.org/POM/4.0.0"
4         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
6                             https://maven.apache.org/xsd/maven-4.0.0.xsd">
7
8     <modelVersion>4.0.0</modelVersion>
9
10    <groupId>com.library</groupId>
11    <artifactId>LibraryManagementExercise4</artifactId>
12    <version>0.0.1-SNAPSHOT</version>
13
14    <properties>
15        <maven.compiler.source>1.8</maven.compiler.source>
16        <maven.compiler.target>1.8</maven.compiler.target>
17    </properties>
18
19    <dependencies>
20
21        <!-- Spring Context -->
22        <dependency>
23            <groupId>org.springframework</groupId>
24            <artifactId>spring-context</artifactId>
25            <version>5.3.30</version>
26        </dependency>
27
28        <!-- Spring AOP -->
29        <dependency>
30            <groupId>org.springframework</groupId>
31            <artifactId>spring-aop</artifactId>
32            <version>5.3.30</version>
33        </dependency>
34    </dependencies>
35</project>
```

view Dependencies Dependency Hierarchy Effective POM pom.xml

```
LibraryManagementExercise4/pom.xml X
30 <groupId>org.springframework.boot</groupId>
31 <artifactId>spring-aop</artifactId>
32 <version>5.3.30</version>
33 </dependency>
34
35 <!-- Spring Web MVC -->
36 <dependency>
37 <groupId>org.springframework</groupId>
38 <artifactId>spring-webmvc</artifactId>
39 <version>5.3.30</version>
40 </dependency>
41
42 </dependencies>
43
44 <build>
45 <plugins>
46
47 <plugin>
48 <groupId>org.apache.maven.plugins</groupId>
49 <artifactId>maven-compiler-plugin</artifactId>
50 <version>3.8.1</version>
51
52 <configuration>
53 <source>1.8</source>
54 <target>1.8</target>
55 </configuration>
56
57 </plugin>
58
59 </plugins>
60 </build>
61
62 </project>
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

Output:

```
WEEK-2 TASKS ECLIPSE - LibraryManagementExercise4/pom.xml - Eclipse IDE
File Edit Navigate Search Project Run Design Window Help

LibraryManagementExercise4/pom.xml X
30 <groupId>org.springframework.boot</groupId>
31 <artifactId>spring-aop</artifactId>
32 <version>5.3.30</version>
33 </dependency>
34
35 <!-- Spring Web MVC -->
36 <dependency>
37 <groupId>org.springframework</groupId>
38 <artifactId>spring-webmvc</artifactId>
39 <version>5.3.30</version>
40 </dependency>
41
42 </dependencies>
43
44 <build>
45 <plugins>
46
47 <plugin>
48 <groupId>org.apache.maven.plugins</groupId>
49 <artifactId>maven-compiler-plugin</artifactId>
50 <version>3.8.1</version>
51
52 <configuration>
53 <source>1.8</source>
54 <target>1.8</target>
55 </configuration>
56
57 </plugin>
58
59 </plugins>
60 </build>
61
62 </project>
```

Overview Dependencies Dependency Hierarchy Effective POM pom.xml

```
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-logger-api/3.2.5/surefire-logger-api-3.2.5.jar (
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-shared-utils/3.2.5/surefire-shared-utils-3.2.5-j
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-api/3.2.5/surefire-extensions-api-3.2
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ LibraryManagementExercise4 ---
[INFO] Building jar: C:\Users\lokal\Downloads\WEEK-2 TASKS ECLIPSE\LibraryManagementExercise4\target\LibraryManagementExercise4-0.0.1-SNAPSHOT
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ LibraryManagementExercise4 ---
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18-jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18-jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18-jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18-jar
[INFO] Installing C:\Users\lokal\Downloads\WEEK-2 TASKS ECLIPSE\LibraryManagementExercise4\target\LibraryManagementExercise4-0.0.1-SNAPSHOT.jar to C:\Users\lokal\m2\repository\com\librar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 16.867 s
[INFO] Finished at: 2026-06-27T13:52:18+05:30
[INFO]
```